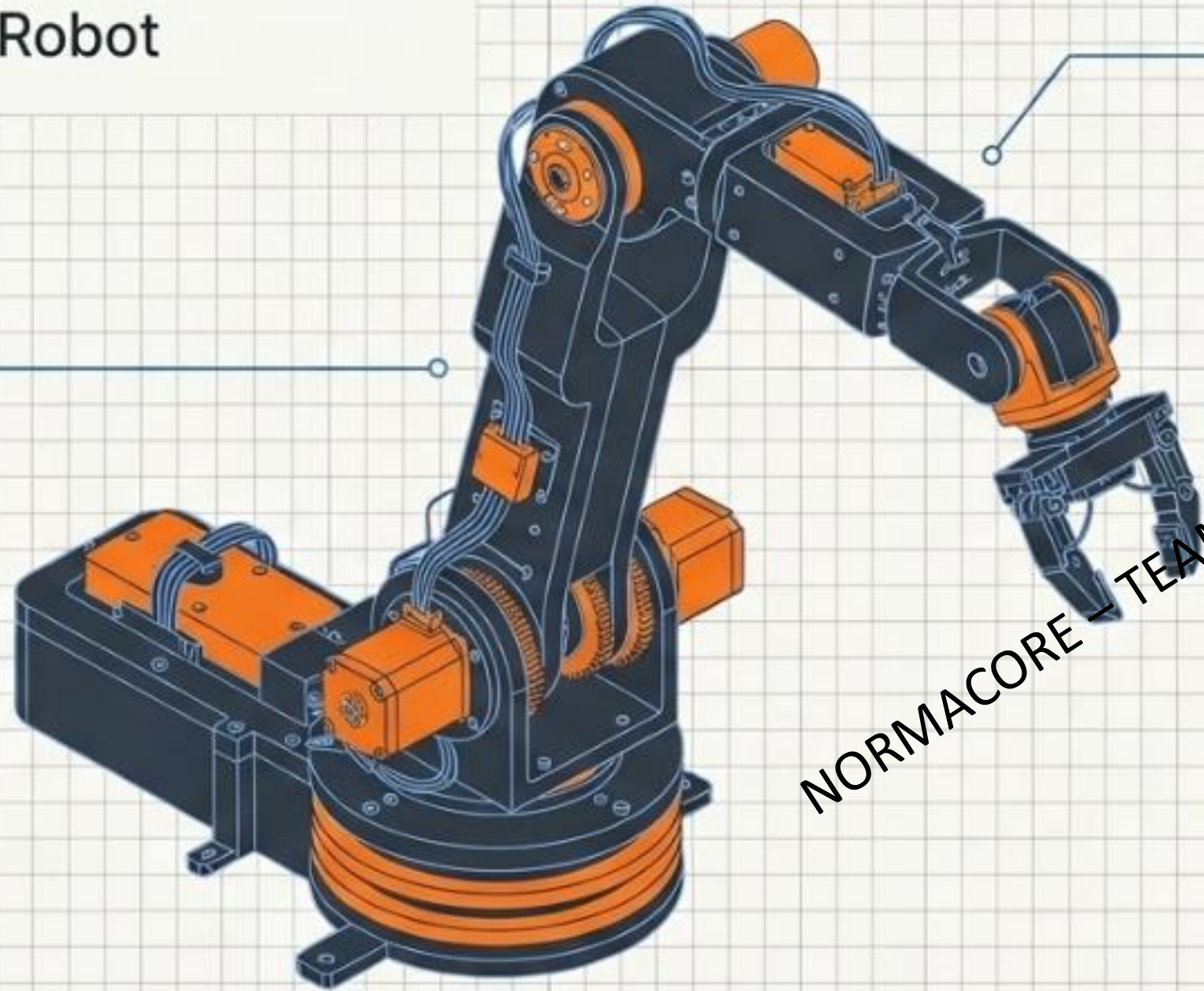


Building an Edge-First AI Robot for \$220

of Voice/Text Control ElRobot

8-DOF
Control

100Hz
Servo Loop



NORMACORE - TEAM 1

#5B61037157

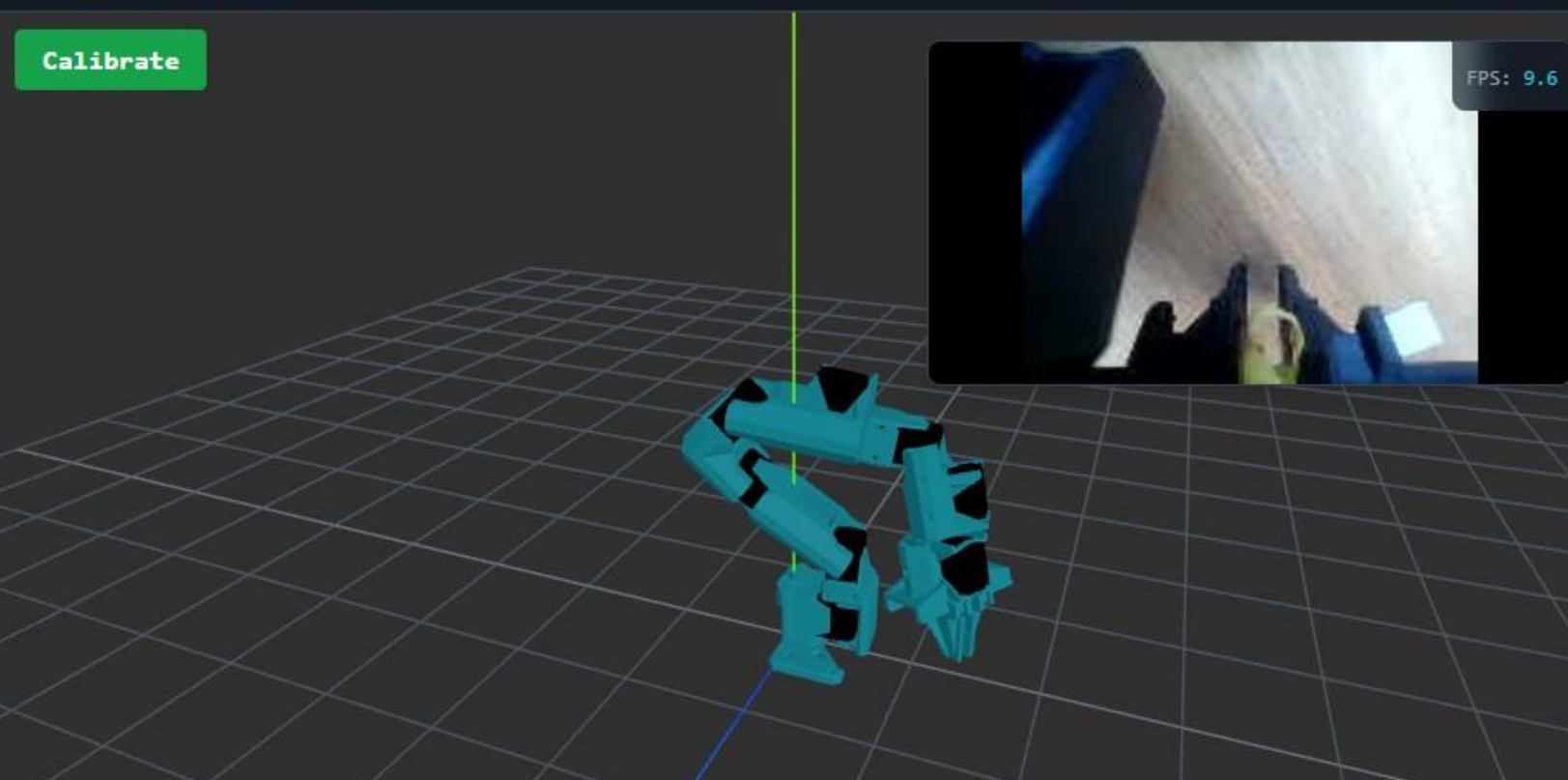
3D

(Self-controlled)

10 (7749:521:3:1)

Port: /dev/ttyACM1 90ms

Calibrate



ID	POS	CURR	RANGE	%	TEMP	LAG	MAX	STATUS	
M1	2110	0	1195	2901	53.6%	33°C	97ms	254ms	OK
M2	1418	0	1011	3085	19.6%	35°C	97ms	248ms	OK
M3	2176	0	0969	3127	55.9%	36°C	98ms	258ms	OK
M4	2461	0	1234	2862	75.4%	39°C	99ms	251ms	OK
M5	1960	0	0182	3914	47.6%	37°C	97ms	252ms	OK
M6	2636	0	1104	2992	81.1%	34°C	98ms	262ms	OK
M7	3483	0	0371	3725	92.8%	35°C	98ms	249ms	OK
M8	1474	0	1277	2819	12.8%	38°C	98ms	253ms	OK

#5B3E089716

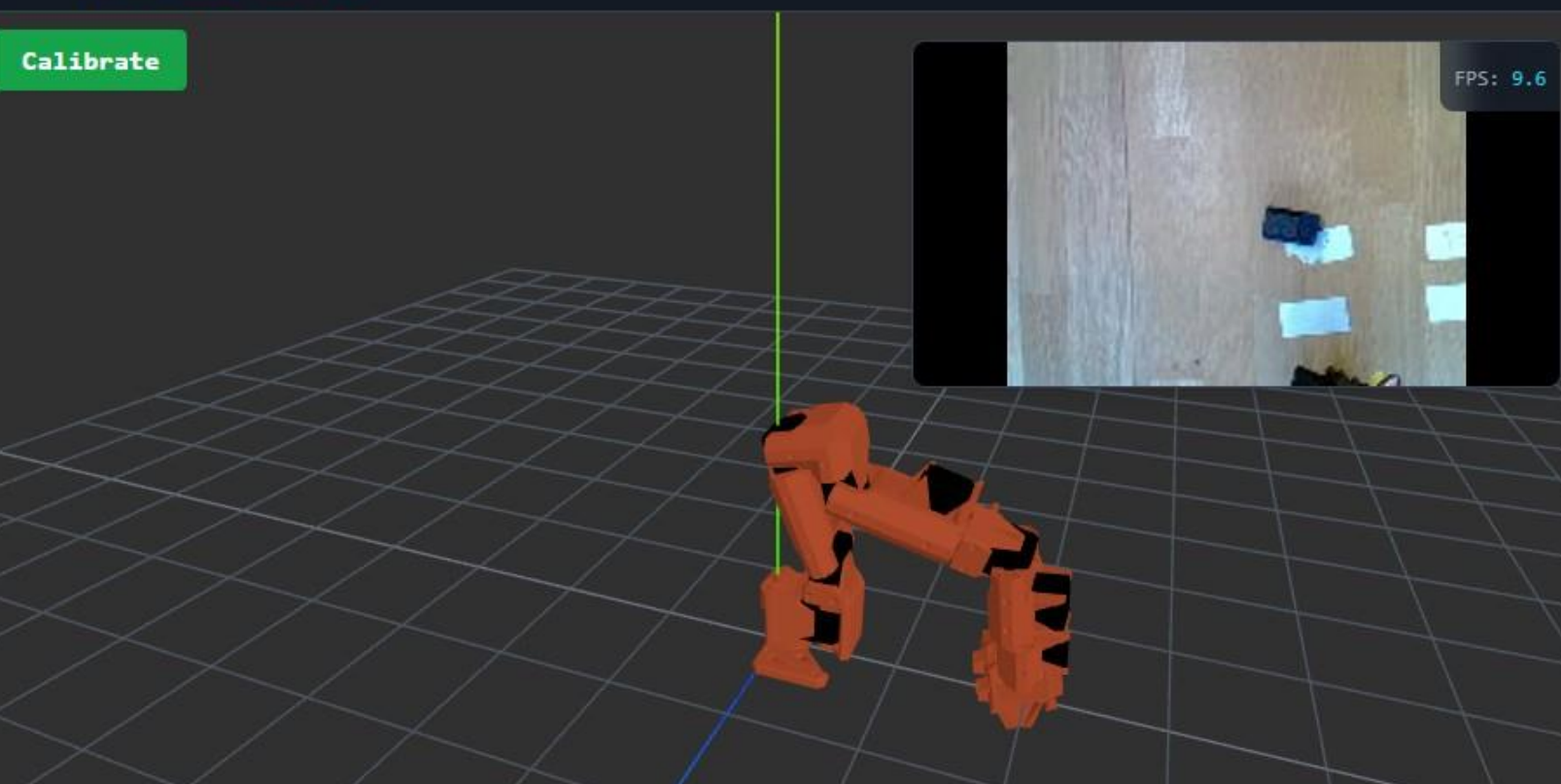
3D

(Self-controlled)

5 (1133:2085:1:5)

Port: /dev/ttyACM2 92ms

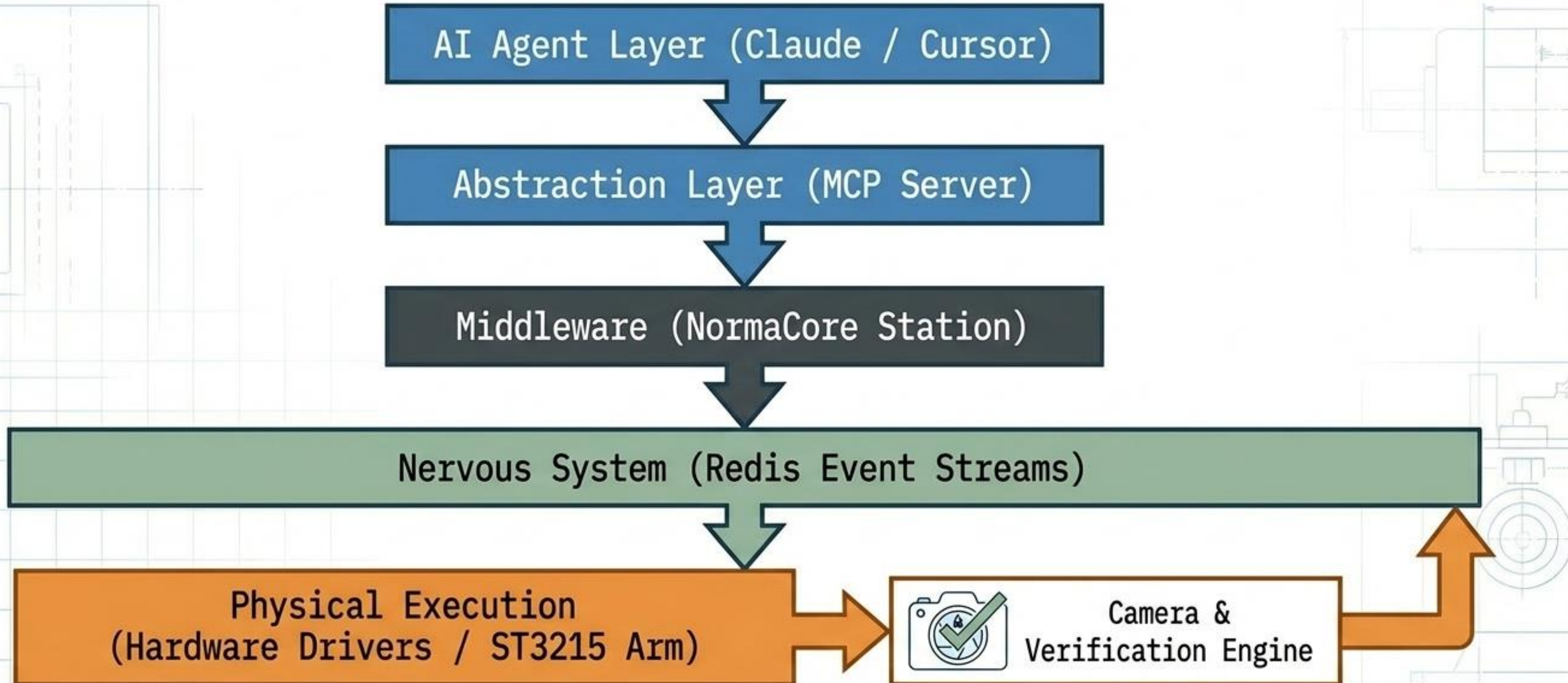
Calibrate



ID	POS	CURR	RANGE	%	TEMP	LAG	MAX	STATUS	
M1	2059	0	1075	3021	50.6%	28°C	100ms	262ms	OK
M2	1797	0	0999	3097	38.0%	29°C	101ms	263ms	OK
M3	3195	0	0936	3160	101.6%	27°C	101ms	258ms	OK
M4	1478	0	1153	2943	18.2%	29°C	100ms	260ms	OK
M5	1754	0	0150	3946	42.3%	27°C	100ms	257ms	OK
M6	2655	0	1046	3050	80.3%	28°C	100ms	259ms	OK
M7	1938	0	1922	2174	6.3%	28°C	101ms	261ms	OK
M8	1871	0	1445	2651	35.3%	29°C	100ms	256ms	OK

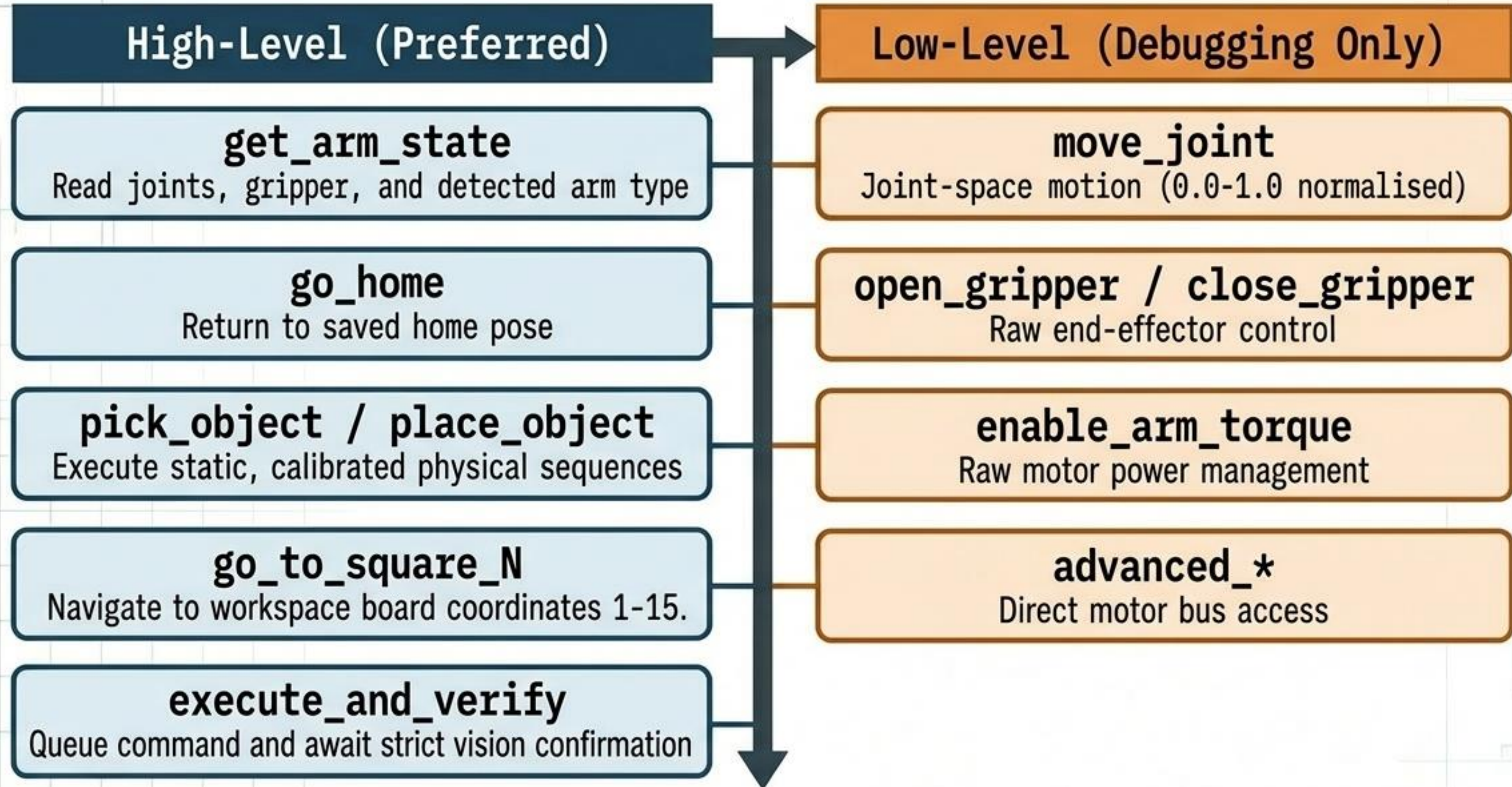
The Top-Down System Architecture

A decoupled pipeline from natural language to motor torque.



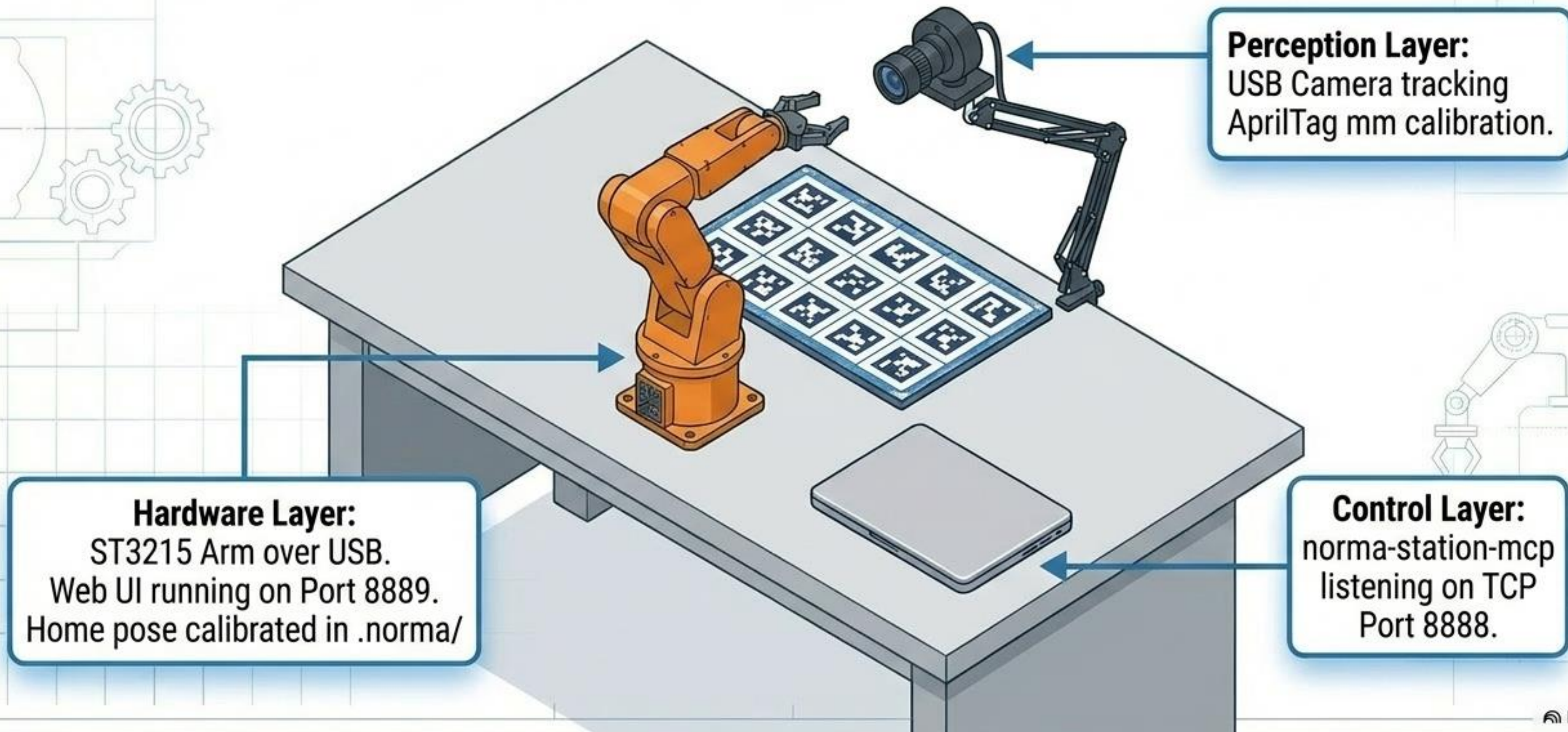
MCP Tooling Taxonomy

Exposing robot capabilities through structured protocol calls.



Phase 2: The Hackathon Build

Deploying the architecture to live physical operations.



Hackathon Capability Dashboard

Interacting with the ST3215 arm via Voice, Vision, and Grid constraints.



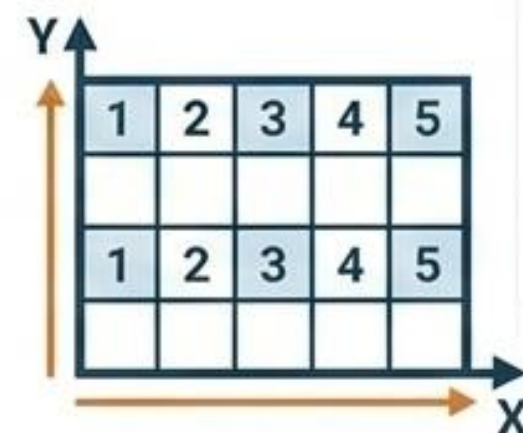
Voice Stack

- Real-time speech input
- Whisper STT to GPT-4o
- Direct MCP Tool Calls
- Persona responses ("Hey Joe")



Vision Stack

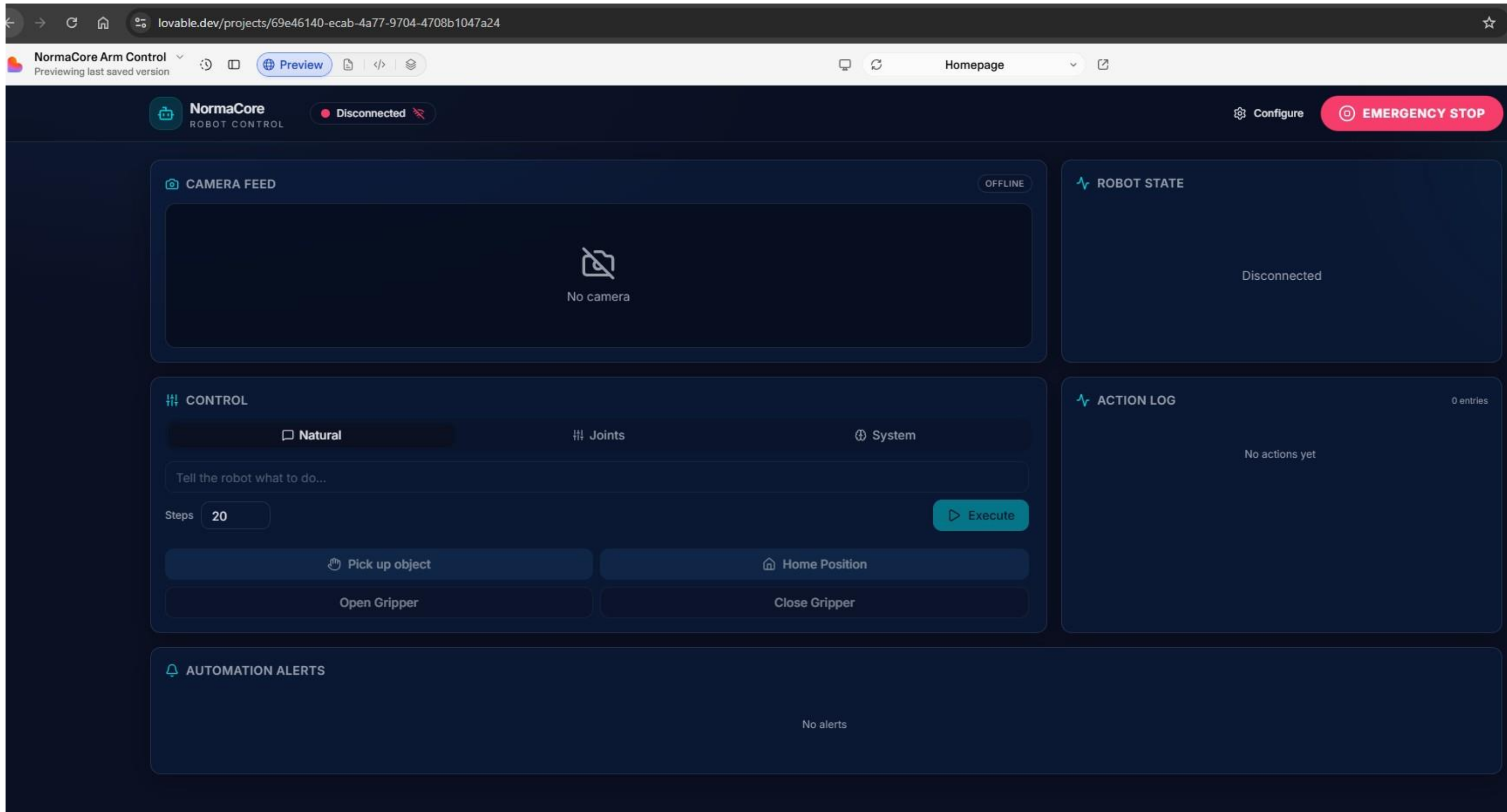
- Browser overlay with contrast detection
- Optional Roboflow API integration
- Python live pipeline for object targeting
- AprilTag workspace mm calibration



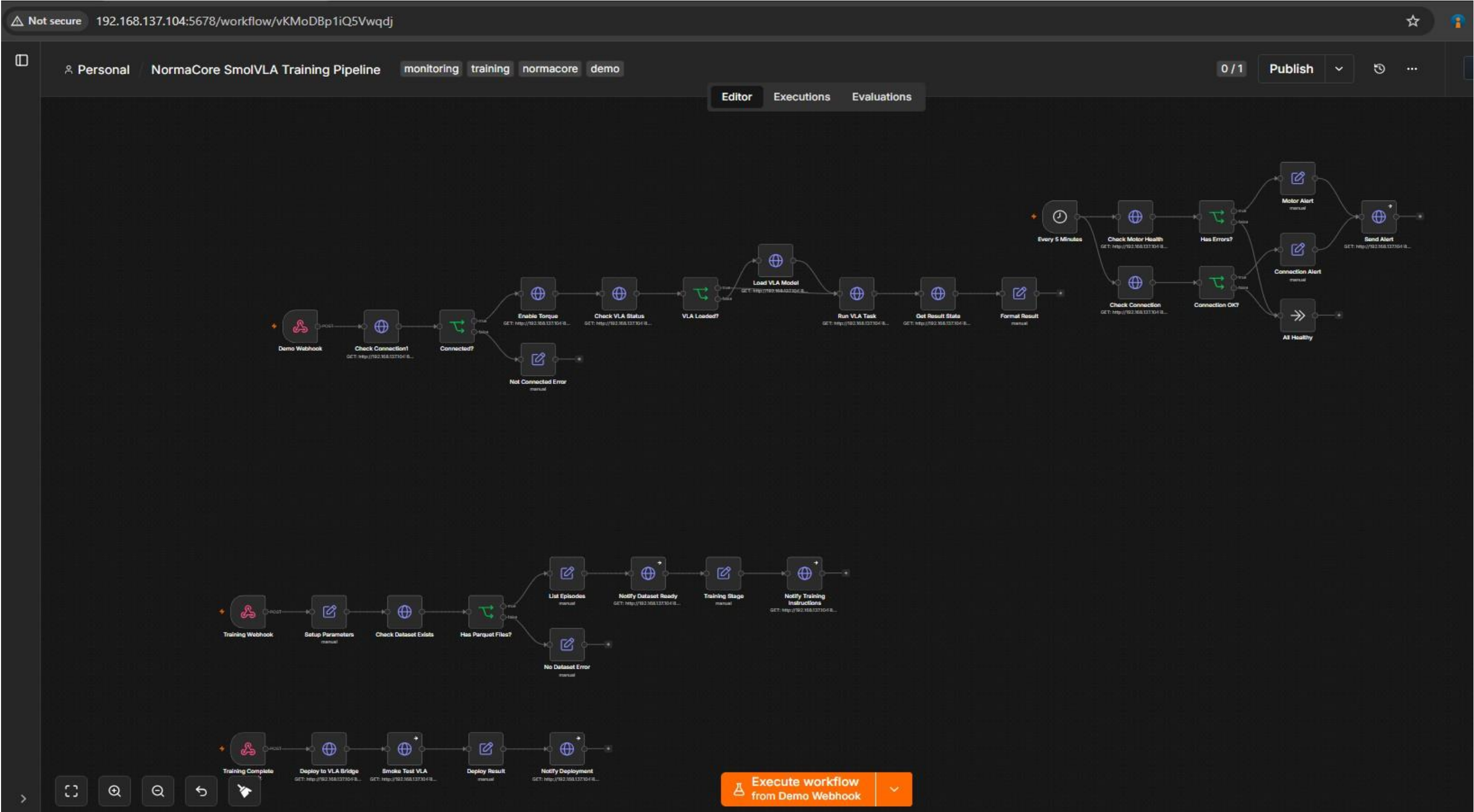
Grid Control

- 5x3 calibrated workspace layout
- Maps AI intent to strict coordinates
- Supports exact grasping (go_to_square_8)
- Calibrated pick and place sequencing

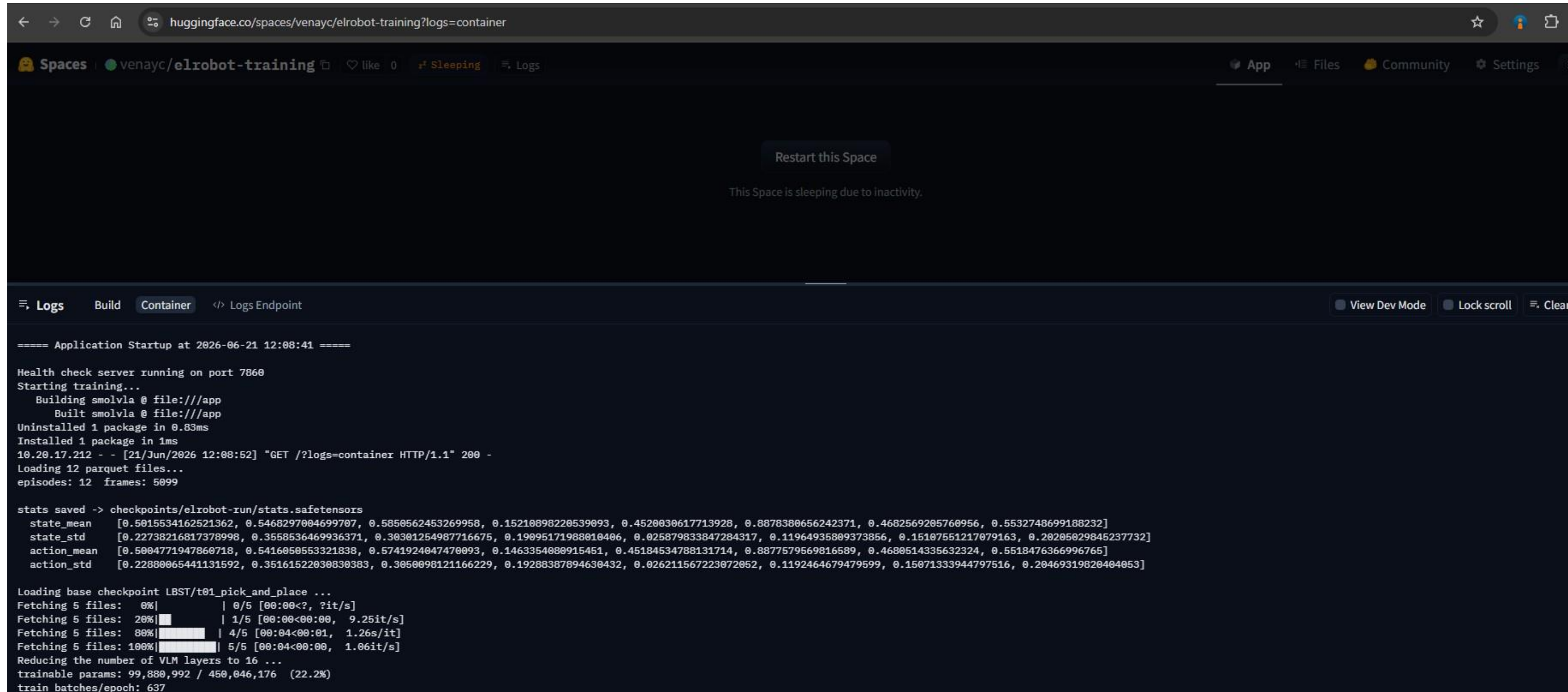
Work - In progress: Lovable Dashboard for Monitoring



Work – In progress : N8N Workflow to automate Health Checkup, Dataset Recorder



Work In-progress : Extensive training of Model in HuggingFace



The screenshot displays the HuggingFace Spaces interface for a Space named 'venayc/elrobot-training'. The interface is in a dark theme. At the top, there's a navigation bar with 'Spaces', 'venayc/elrobot-training', 'like 0', 'Sleeping', and 'Logs'. Below this, a large button labeled 'Restart this Space' is visible, with a message 'This Space is sleeping due to inactivity.' underneath it. The main content area shows the 'Logs' tab selected, displaying a detailed log of the application startup and training process. The log includes information about the application startup, health check server, training start, package management, and the loading of 12 parquet files. It also shows the saving of stats to a file and the loading of a base checkpoint. The log ends with the number of trainable parameters and the number of training batches/epochs.

```
===== Application Startup at 2026-06-21 12:08:41 =====  
  
Health check server running on port 7860  
Starting training...  
  Building smolvla @ file:///app  
  Built smolvla @ file:///app  
Uninstalled 1 package in 0.83ms  
Installed 1 package in 1ms  
10.20.17.212 - - [21/Jun/2026 12:08:52] "GET /?logs=container HTTP/1.1" 200 -  
Loading 12 parquet files...  
episodes: 12  frames: 5099  
  
stats saved -> checkpoints/elrobot-run/stats.safetensors  
state_mean  [0.5015534162521362, 0.5468297004699707, 0.5850562453269958, 0.15210898220539093, 0.4520030617713928, 0.8878380656242371, 0.4682569205760956, 0.5532748699188232]  
state_std   [0.22738216817378998, 0.3558536469936371, 0.30301254987716675, 0.19095171988010406, 0.025879833847284317, 0.11964935809373856, 0.15107551217079163, 0.20205029845237732]  
action_mean [0.5004771947860718, 0.5416050553321838, 0.5741924047470093, 0.1463354080915451, 0.45184534788131714, 0.8877579569816589, 0.4680514335632324, 0.5518476366996765]  
action_std  [0.22880065441131592, 0.35161522030830383, 0.3050098121166229, 0.19288387894630432, 0.026211567223072052, 0.1192464679479599, 0.15071333944797516, 0.20469319820404053]  
  
Loading base checkpoint LBST/t01_pick_and_place ...  
Fetching 5 files: 0%|          | 0/5 [00:00<?, ?it/s]  
Fetching 5 files: 20%|██      | 1/5 [00:00<00:00, 9.25it/s]  
Fetching 5 files: 80%|██████  | 4/5 [00:04<00:01, 1.26s/it]  
Fetching 5 files: 100%|████████| 5/5 [00:04<00:00, 1.06it/s]  
Reducing the number of VLM layers to 16 ...  
trainable params: 99,880,992 / 450,046,176 (22.2%)  
train batches/epoch: 637
```